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SYSTEMATIC REVIEW

Examination of Reaction Time Deficits Following Concussion: A Systematic Review and Meta-analysis

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Abstract

Background Reaction time (RT) deficits are reported following concussion, but it is unknown when these deficits normalize to pre-injury status. It is also unclear how factors such as RT measurement technique and participant characteristics influence post-concussion RT.

Objective The purpose of this systematic review and meta-analysis was to (1) characterize acute post-concussion (0–3 days) RT impairments, (2) examine RT recovery over time, and (3) explore moderating factors related to acute RT impairment following concussion.

Methods Database searches (PubMed, CINAHL, EBSCOhost) were conducted according to PRISMA guidelines for articles published in English from January 2002 to March 2019. Studies compared baseline-to-post-injury RT within individuals (within-subject) and/or RT in concussed individuals to non-concussed controls (between-subject). Sixty studies met inclusion criteria, reporting on a total of 9688 participants with 214 discrete RT effects (Hedges' *d*; between-subject: $N=29$, $k=129$; within-subject: $N=42$, $k=85$). Of the 214 effects, 93 occurred in the acute (0–3 days) post-injury timeframe ($k=47$ between-subject). Numerous demographic [sex, age, concussion history, population type (athlete, military, and general population), athlete level (high school, college), and sport], and method-based (RT test and measure type, computerized neurocognitive testing platform, concussion definition, and time post-injury) moderators were examined for mean effect influence. Mixed-effects multi-level modeling with restricted-maximum-likelihood estimation was used to account for nested effects and high heterogeneity for the pooled effect size ($D+$).

Results Significant medium-magnitude RT deficits were observed acutely for between- ($D+=-0.7279$, 95% CI -0.9919, -0.4639, $I^2=88.66$, $p<0.0001$) and within-subject ($D+=-0.7472$, 95% CI -0.9089, -0.5855, $I^2=89.21$, $p<0.0001$) effect models. RT deficits were present at the sub-acute and intermediate-term timeframes for between-subject effects (sub-acute: $D+=-0.5655$, 95% CI -0.6958, -0.4352, $p<0.0001$; intermediate-term: $D+=-0.3219$, 95% CI -0.5988, -0.0450, $p=0.0245$). No significant RT mean effect was observed for the between-subject model at the long-term timeframe, indicating RT recovery among concussed participants relative to controls ($D+=0.3505$, 95% CI -0.4787, 1.1797, $p=0.3639$). Sex was a significant moderator for between-subject effects, with every 1% male sample size increase demonstrating -0.0171 (95% CI -0.0312, -0.0029, $p=0.0193$) larger RT deficits. Within-subject effect models resulted in RT measure type (simple: [$D+=-0.9826$] vs. mixed: [$D+=-0.6557$], $p=0.0438$) and computerized neurocognitive testing platforms (ANAM: [$D+=-0.3735$] vs. HeadMinder CRI: [$D+=-1.4799$] vs. ImPACT: [$D+=-0.6749$], $p=0.0004$) having significantly different RT-deficit magnitudes. No other moderators produced significantly different RT-deficit magnitudes (between-subject: [$p\geq0.0763$], within-subject: [$p\geq0.1723$]).

Conclusions Robust RT deficits were observed acutely following concussion. Minimal magnitude differences were noted when comparing between- and within-subject effects, suggesting that pre-injury baselines may not add clinical value in determining post-injury RT impairment. RT deficits persisted up till the intermediate-term (21–59 days post-injury) timeframe and indicate lingering deficits exist. Mean effect size differences were observed between RT measure types and computerized neurocognitive testing platforms; however, all categories displayed negative effects consistent with impaired RT following concussion. Clinical interpretation suggests that measuring RT post-concussion is more important than considering the RT method employed so long as reliable and valid tools are used. PROSPERO Registration #CRD42019119323.

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Extended author information available on the last page of the article

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Key Points

Robust reaction time deficits are present in the first 3 days following concussion.

Post-concussion reaction time deficits persist until around 59 days post-injury and indicate lingering impairment not reported in individual studies.

Population, methodological, and reaction time assessment characteristics are statistically significant, but clinically limited, moderators for post-concussion reaction time deficits.

1 Introduction

Concussion is broadly defined as a "traumatic brain injury induced by biomechanical forces" [1], and is common among athletic [2, 3], military [4], and the general public [5]. Current consensus statements recommend using a multidimensional concussion assessment battery including concussion signs and symptoms, balance, and neurocognitive function at minimum to provide the strongest diagnostic accuracy [1, 6, 7]. Computerized neurocognitive assessments measuring numerous psychometric domains are commonly employed in athletic- [8, 9], military- [10, 11], and hospital-based [12] medical settings. Reaction time (RT) is frequently measured via computerized neurocognitive assessments [13, 14] and is among the most reliable and robust neurocognitive domains for post-concussion assessments [15].

Reaction time is critical to sport performance and everyday life. For example, athletes must respond quickly to a ball or another athlete's movement to successfully compete and avoid injury [16]. Drivers must step on a brake pedal if another vehicle pulls out in front to avoid a collision [17]. Thus, RT is fundamental to numerous activities across populations and is considered a critical post-concussion outcome. Post-concussion RT deficits are frequently reported in the literature [13, 18, 19], with RT outcomes moderately predicting symptom recovery [20]. Despite previous work, post-concussion RT recovery varies across studies, making it unclear when RT returns to pre-injury baseline status.

Different methods for measuring RT may contribute to variances in reported RT recovery following concussion. Reaction time has traditionally been assessed through computerized neurocognitive testing; however, relatively more simplistic (e.g., manually catching a falling measuring stick) [18, 21] and advanced (e.g., full-body motor responses to a visual stimulus) RT assessments have emerged [22]. Computerized neurocognitive assessments commonly measure RT using a computer keyboard or mouse (i.e., single finger movement) in response to a stimulus displayed on a computer monitor, and may differ greatly when compared to lower extremity [23] or whole-body movement-based [19, 22] RT measurements such as those required in sport and activities of daily living. The numerous clinically available computerized neurocognitive platforms also incorporate different RT measures often categorized as simple, procedural, or mixed. It is important to note the vast majority of post-concussion studies are technically calculating response time (i.e., total time from stimulus presentation to movement completion) which is the sum of reaction time (i.e., time from stimulus presentation to movement initiation) and movement time (i.e., time from movement initiation to movement completion) [16]. Response time is what clinicians and researchers focus on as it represents the entire neurocognitive and motor output time needed to complete the task [13, 18, 22]. However, because it is standard terminology in the concussion literature, we use the term "reaction time" throughout this manuscript.

Simple RT involves responding to a single stimulus with a single response; procedural (sometimes noted as choice or complex RT) involves either multiple stimuli, multiple response options, or both; and mixed refers to a composite RT score derived from combinations of RT measure types. Additionally, differences in study design (within-subject versus between-subject) may influence post-concussion RT findings due to the different comparison group used. Populations studied, such as athletic, general public, and military personnel, may also contribute to contrasting findings. Studies performed among athletes have primarily focused on American Football and soccer in male populations, further questioning the generalizability of the cumulative findings. Finally, the RT calculation method (e.g., raw time measurements [18] versus domain score formulations [24]) may further affect reported outcomes.

These numerous issues pose a problem for researchers and clinicians, because it remains unknown how robust RT deficits are, when these deficits return to pre-injury status, and whether methodological variability affects post-concussion RT results. Therefore, the purpose of this systematic review and meta-analysis is to (1) characterize acute post-concussion (0–3 days) RT impairments, (2) examine RT recovery over time following concussion, and (3) explore moderating factors related to acute RT impairment following concussion.

2 Methods

This systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines [25] and registered a priori with the PROSPERO database [26] (Registration # CRD42019119323). Articles published between January 1st, 2002 and March 8th, 2019 were retrieved from PubMed, EBSCOhost, and CINAHL using the following search terms:

> "(concuss* OR "mild traumatic brain injury" OR "mtbi" OR "mild TBI" OR "mild brain injury") AND ("reaction time" OR "response time" OR "latency" OR "ImPACT" OR "Immediate Post-Concussion And Cognitive Test" OR "drop stick" OR Clinical reaction time OR "CNS Vital Signs" OR "Head-Minder" OR "ANAM" OR "neurocognitive test*" OR "neuropsychological test*" OR "CogState" OR "Axon" OR "CogSport") NOT (review OR systematic review)"

All duplicate articles identified from the database searches were removed using reference management software.

2.1 Study Selection

Studies meeting the following criteria were included: (1) peer-reviewed publication, (2) published in English, (3) published between 2002 and 2019, (4) involved only human subjects, (5) compared RT in concussed individuals to their own pre-injury baseline (within-subject design) or between concussed individuals and a non-concussed control group (between-subject design), and (6) concussion was operationally defined using either a published definition describing altered mental status following trauma to the head or body, or as being diagnosed by a healthcare professional. The publication year range began at 2002, because the first international consensus statement on concussion in sport was published then and provided a foundational international operational definition of concussion [27]. Exclusion criteria were the following: (1) gray literature sources (e.g., conference papers, abstracts, theses, and dissertations), (2) case study or case-series design, (3) inability to distinguish cases of moderate or severe traumatic brain injuries from the concussed sample, (4) post-injury assessment(s) were not completed < 14 days, and (5) comparison of concussed subjects only to normative data. Studies only comparing concussed subjects to normative data were excluded in line with Cochrane guidelines [28] due to the often unknown normative data participant characteristics and risk for participants in the normative database to be utilized across multiple studies.

The study selection process is outlined in Fig. 1. The initial search identified 5156 studies (3881 following duplicate removal). Articles were split evenly into two blocks that were independently reviewed by two members of the author team (LBL and RCL). During the initial title, abstract, and full-text review, the same two authors reviewed the same 10% study sample to determine reliability throughout the study selection process. Cohen's kappa (κ) coefficient values [29] and 95% confidence intervals (CI) for this reliability assessment are reported in Fig. 1. The authors of studies ($n = 29$) lacking statistical output presented in numeric format for effect size calculations were contacted up to two times via email to request the missing information. Authors were given 4 weeks to provide the requested mean and standard deviation values prior to study exclusion. Of these 29 author groups, 9 (31.0%) provided the requested statistical output (Fig. 1). Following full-text review and author group statistical requests, a final set of 60 studies were identified for quantitative review. One author (LBL) then extracted all study variables during the final quantitative review.

2.2 Effect Size Calculation

Effect sizes were calculated as Hedges' d [30], a variance-adjusted calculation of Hedges' g, to adjust for small-sample bias based on the study-provided means and standard deviations of either pre-injury relative to post-injury (within-subject design) or control group relative to concussed group (between-subject design) values. Effect sizes were calculated using the Metafor package [31] in the R Project for Statistical Programming (v 3.4.3, Murray Hills, NJ, USA) [32]. All effect sizes were calculated, such that slower RT (i.e., an increase in RT; worse performance) post-injury compared to baseline, or in the concussed group relative to controls, resulted in a negative effect size. Multiple effects were calculated for each study due to numerous time points and multiple RT tests employed if applicable. All effect sizes were given a unique effect size identifier nested within an identifier unique to each study.

2.3 Selection of Moderators

The following potential moderators of cognitive function, concussion recovery, and/or RT [33] were extracted from each study if reported: time to assessment post-injury, sex (% male), mean age in years, concussion history (% sample and mean frequency), population type (athlete, military, general population), athlete playing level (e.g., high school, collegiate), sport sample (% football and % soccer), concussion definition used, RT test type (clinical, computerized neurocognitive testing, and "other"), RT measure type (e.g. simple, procedural, mixed), and

computerized neurocognitive testing platform [e.g., Automated Neuropsychological Assessment Metrics (ANAM), HeadMinder Concussion Resolution Index (HeadMinder CRI), and Immediate Post-Concussion Assessment and Cognitive Test (ImPACT)]. At least five individual effect sizes for each moderator level were required for inclusion in the moderator analyses [34].

Since most patients recover from concussion on clinical measures within 14 days [7], elapsed time between injury and assessment may influence cumulative effect findings. To control for time from injury, effect sizes were binned into the following time periods: acute (0–3 days), sub-acute (4–17 days), intermediate term (21–59 days), and long term (80–365 days). Time between assessment and injury was considered as a continuous variable, but time points were not normally distributed and were heavily weighted to 0–3 days post-injury. Reaction time test types were classified as "clinical" if the test did not require computer hardware/software and needed minimal equipment [18, 35], as "computerized neurocognitive testing" if an established platform was used [8, 10, 13, 36], and as "other" (e.g., pressing a button during gait [37] or turning a steering wheel [38]) if they did not fit into "clinical" or "computerized neurocognitive testing".

Numerous RT measurement types are described in the literature and nomenclature differs widely between disciplines and assessment tools. In the present study, RT types were classified as simple, procedural, or mixed based on either the classification used by the testing platform or the classification described in the source study.

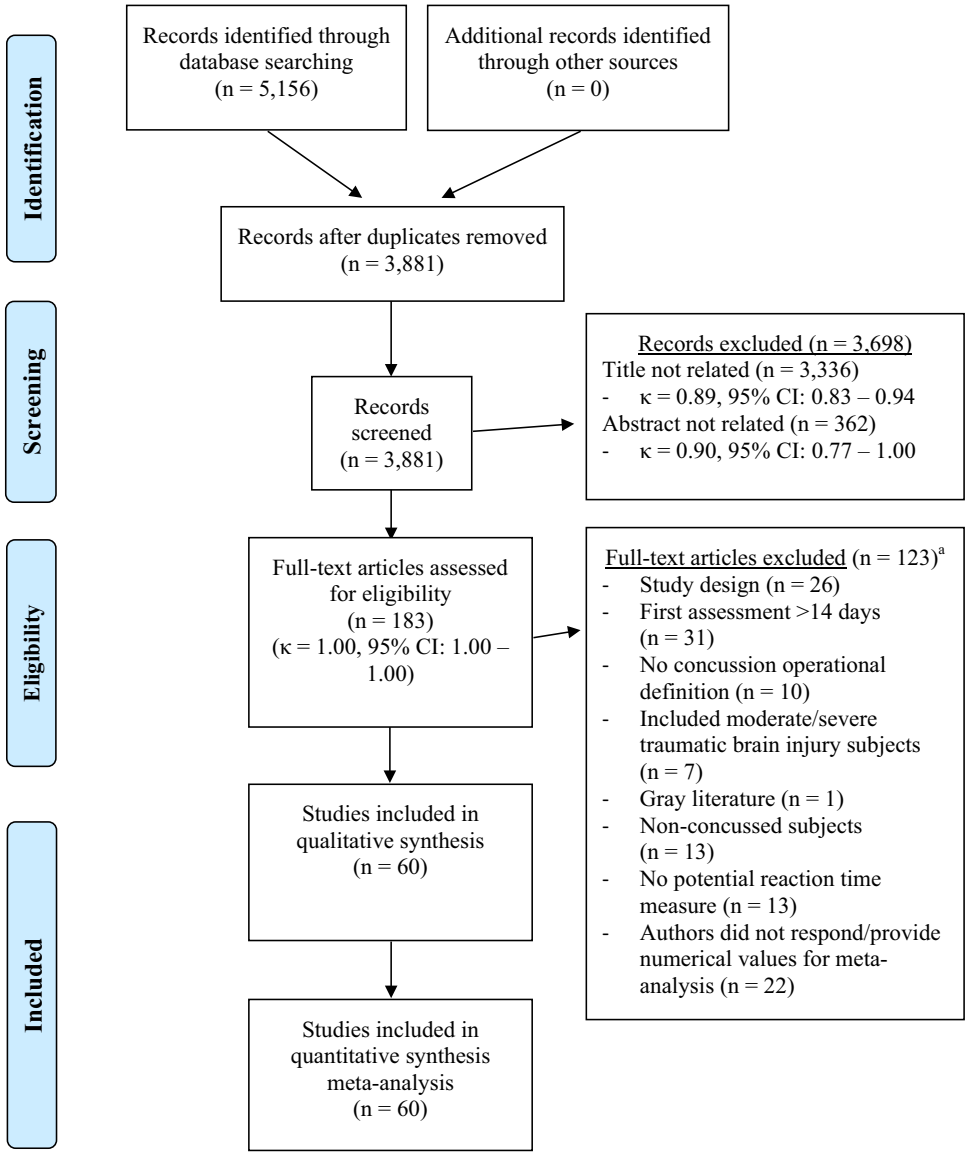

Fig. 1 PRISMA flowchart. [a]Articles identified for exclusion were based on the first identified exclusion criterion listed. Studies may have met multiple exclusion criteria

2.4 Study Quality Assessment

The Newcastle–Ottawa Scale (NOS) was used to assess study quality [39]. The NOS is a reliable quality assessment tool for evaluating observational studies recommended by the Cochrane Collaboration group [28, 40]. All articles included for quantitative analysis were split into two blocks and independently scored by two study team members (RCL and LBL) on the 0–9 NOS (9 is the highest possible score). A 10% study sample ($n = 8$) was scored by both authors to establish intraclass correlation coefficient [41] (ICC) NOS agreement, with strong agreement present ($ICC_{3,1} = 0.91$, 95% CI 0.60–0.98).

2.5 Statistical Analysis

Due to the inherent differences in comparison groups for between- and within-subject study designs, effects were analyzed separately for the two study designs. Random-effects models with restricted maximum-likelihood estimation were initially used to aggregate the standardized mean differences for between- and within-subject effects separately. Effect size heterogeneity was tested using Q and I^2 statistics.[42] A significant Q value indicates heterogeneity between study effects, whereas I^2 indicates the percentage of between-effect variance not attributable to sampling error. I^2 values are typically interpreted as insignificant (< 25%), low (26–50%), moderate (51–75%), or high (> 75%) heterogeneity [42].

All moderator analyses were pre-specified to avoid spurious non-theoretical findings [28]. Moderator analyses were limited to acute (0–3 days post-injury) assessment effects, with the exception of time post-injury to examine RT recovery via binned time points, to reduce heterogeneity and clarify moderator relationships. Mixed-effects multi-level linear regression models with restricted maximum-likelihood estimation were used to account for the high heterogeneity in both the between- ($Q_{46} = 465.4981$, $I^2 = 92.54\%$, $p < 0.0001$) and within-subject ($Q_{45} = 295.9764$, $I^2 = 93.39\%$, $p < 0.0001$) acute random-effects models, as well as the presence of multiple nested effects within studies [43, 44]. Only RT recovery analyses were conducted (separately for between- and within-subject) on all post-injury time point effect sizes with time between injury as the moderating effect to examine RT recovery. Pooled mean effect sizes with associated 95% CI were calculated and interpreted according to conventional statistical guides with ≤ 0.20, 0.21–0.79, and ≥ 0.80 used as thresholds for small, medium, and large effects, respectively [45]. Moderator analyses were conducted on continuous and categorical moderators and differ slightly in interpretation. Similar to linear regression models, continuous moderator mean effect sizes are interpreted as every one-unit increase in the moderator alters the mean effect size by the corresponding value. Categorical moderators are interpreted similar to ANOVA models, such that an omnibus model is tested for differences between the groups, and the effect size differences between groups being the primary focus. All effect sizes with 95% CI not including zero, or not overlapping for groups among categorical moderators, were considered statistically significant.

The number of unpublished effects averaging zero (i.e., null effects) not retrieved from studies of similar sample size that would reduce the significance of observed effects to $p > 0.05$ was estimated using a fail-safe N+ method [46]. Finally, Q–Q plots were used to assess normality and funnel plots were graphed to assess for potential publication bias, with Egger's test statistic calculated to quantitatively assess funnel symmetry [47, 48]. Statistical analyses were conducted using the Metafor package [31] for the R Project for Statistical Programming (v 3.4.3, Murray Hills, NJ, USA) [32].

3 Results

A total of 214 effects were present across the 60 included studies. Of the 60 included studies, a total of 129 between-subject effects were present across 29 studies and 85 within-subject effects were present across 42 studies. The number of effects exceeds the number of studies, because numerous studies used multiple RT measures (between-subject: 15, within-subject: 9), multiple post-injury time points (between-subject: 20, within-subject: 22), and/or included both between- and within-subject effects (mixed study design; 12). Of the 214 effects, 93 occurred in the acute (0–3 days) post-injury timeframe (47 between-subject). The 60 studies included a total of 9688 subjects (between-subject: 3308) with 7161 having concussions (between-subject: 1825). Tables 1 and 2 summarize the study and sample characteristics. Electronic Supplementary Material Tables S1 and S2 further describe the measurement tools, outcome measures, and NOS scores.

3.1 Methodological Quality and Effect Size Assessments

Fail-safe N+ calculations indicated that additional 5669 between-subject studies and 14,565 within-subject studies of similar sample size with non-significant effects would be needed to decrease the mean effect significance to $p > 0.05$. The Q–Q and funnel plots presented in Electronic Supplementary Material Figure S1 depict the heterogeneity of study effects. Egger's test (Electronic Supplementary Material Figure S1) indicated statistically significant asymmetries for between-subject ($z = -4.7533$, $p < 0.0001$) and within-subject studies ($z = -6.0697$, $p < 0.0001$) consistent with heterogeneity among effects and/or potential publication

bias. Between- and within-subject studies had acceptable average NOS scores of 7.41 and 7.56, respectively.

3.2 Mean Effects

Random-effects pooled study effect sizes from the acute (0–3 days) post-injury time point are presented in Fig. 2 (between-subject) and Fig. 3 (within-subject) with individual study, pooled mean effects ($D+$), and associated 95% CI. Significant medium-to-large mean effects using random-effects modeling were present for both between-subject ($D+=-0.7170$, 95% CI -0.9438, -0.4902, $Q_{46}=465.4981$, $I^2=92.54\%$, $p<0.0001$) and within-subject ($D+=-0.8511$, 95% CI -1.0321, -0.6700, $Q_{45}=295.9764$, $I^2=93.39\%$, $p<0.0001$) designs, indicating that large RT deficits are present acutely following concussion. Results from the mixed-effects multi-level model demonstrated similar but reduced RT deficits after controlling for nested effects within studies (between-subject: $D+=-0.7279$, 95% CI -0.9919, -0.4639, $p<0.0001$; within-subject: $D+=-0.7472$, 95% CI -0.9089, -0.5855, $p<0.0001$; Table 3).

3.3 Moderator Analyses

Moderator analyses examining when RT deficits return to pre-injury status were independently conducted on all the between- ($k=129$) and within-subject ($k=85$) effects separately over the binned time points. All other moderator analyses conducted were restricted to the acute (0–3 days post-injury) timeframe. Mixed-effects multi-level models were employed to control for numerous effects from a single study. The between-subject analysis examining effects over time indicated significant pooled effects for acute, sub-acute, and intermediate-term post-injury time frames ($D+\geq-0.7279$, $p\leq0.0245$; Table 3). All timeframes indicated medium negative pooled effects except for long term ($D+=0.3505$, 95% CI -0.4787, 1.1797), suggesting that consistent RT deficits may linger up to 59 days post-injury. The within-subject analysis only examined acute and sub-acute time points due to limited effects at intermediate term ($k=3$) and long term ($k=0$). Significant pooled effects for acute and sub-acute time points were present ($D+\geq-0.7472$, $p<0.0001$; Table 3), with acute displaying a medium pooled effect and sub-acute displaying a small pooled effect.

Moderator analyses on acute time point between- ($k=47$) and within-subject ($k=46$) effects revealed that sex (% male) was the only independent moderator of acute RT deficits following concussion for between-subject effects (Table 3). Within-subject moderator analyses indicated that the RT measure type (simple vs. mixed) and computerized neurocognitive test type (ANAM, HeadMinder CRI, and ImPACT) were the only independent moderators (Table 3). No other moderators were statistically significant for between- ($p\geq0.0763$) or within-subject effects ($p\geq0.2853$).

4 Discussion

Our study provides novel insight into acute RT deficits, when RT recovery occurs, and factors that influence acute RT deficits following concussion. Overall, medium-to-large-effect RT deficits were observed across multiple models, suggesting robust acute post-concussion impairments. We found methodological differences such as study design (between-subject vs within-subject), sex, RT measure type, and computerized neurocognitive testing platform to statistically influence RT deficits. However, mean effect sizes for post-concussion RT deficits remained small-to-large ($D+$ range -0.3735, -0.7472) when controlling for these methodological factors. As such, these methodological differences appear minimal relative to the robust effects of concussion on RT.

We observed overall RT deficits that extend well beyond the traditional clinical recovery timeframe. Between-subject effects indicated that significant RT deficits were still present during the intermediate-term (21–59 days post-injury; $D+=-0.3219$), but resolved and improved by the long-term period (80–365 days post-injury; $D+=0.3505$). Effect sizes were binned based on time between injury and post-injury assessment, which could influence mean effects if individual effect size distributions were heavily weighted towards earlier or later days within a post-injury category. However, 13 [49–51] of the 26 between-subject effects at the intermediate term occurred between 26–30 days and the remaining 13 [49, 50, 52–55] occurred between 43–59 days, indicating relatively equal effect size distributions in this timeframe. Large-scale longitudinal and epidemiological studies have reported clinical recovery to occur for the vast majority of high-school and collegiate athletes within 28 days [7, 56, 57]; however, our findings indicate that there may be lingering RT deficits that extend beyond clinical recovery. Our findings suggest RT likely has not truly recovered at the time individuals' return to sport, military service, or activities of daily living.

Observing post-concussion changes beyond clinical recovery is not an uncommon finding among emerging dual-task [19, 58] and advanced neuroimaging/biomarker studies [59–61]. One likely reason for extended RT deficits is that we examined the cumulative magnitude of group differences (i.e., effect sizes) rather than examining the probability of differences between groups due to chance (i.e., p values). The included research studies often reported p values, but did not always report effect sizes, one of the few statistical outcomes minimally influenced by sample size

Table 1 Between-subject study and sample characteristics

Author	Sample size (% concussed)	Sex (% male)	Age mean (SD)	Concussion history mean (SD)	Concussion history (%)	Population	Athlete level	American Football (%)	Soccer (%)	Concussion definition used	NOS
Broglio et al. [49]	45 (53.33%)	80.00	16.70 (2.50)	1.01 (1.10)	–	ATH	HS	–	–	NR	8
Catena et al. [37]	28 (50.00%)	57.14	22.29 (3.76)	– (–)	35.71	GEN	–	–	–	AAN	8
Cole et al. [99]	208 (39.90%)	85.54	31.45 (7.14)	1.20 (1.64)	–	MIL	–	–	–	DoD	8
Collie et al. [36]	145 (42.07%)	100.00	23.19 (3.67)	2.29 (2.24)	–	ATH	O	0.00	0.00	ICISG	6
Eckner et al. [18]	54 (48.15%)	92.31	– (–)	– (–)	–	ATH	C	74.07	7.41	NR	9
Fazio et al. [84]	192 (63.54%)	68.75	16.90 (–)	0.27 (0.64)	18.75	ATH	B	43.23	15.63	AAN	6
Gardner et al. [85]	92 (55.43%)	100.00	24.24 (3.94)	2.40 (2.15)	–	ATH	O	0.00	0.00	WHO	6
Hammeke et al. [52]	24 (50.00%)	100.00	16.50 (0.52)	– (–)	–	ATH	HS	100.00	–	AAN	8
Haran et al. [77]	528 (83.33%)	100.00	29.10 (5.00)	– (–)	–	MIL	–	–	–	DoD	6
Heitger et al. [38]	74 (50.00%)	64.86	29.15 (12.65)	– (–)	–	GEN	–	–	–	NR	9
Howell et al. [53]	59 (30.51%)	64.41	19.27 (0.99)	0.71 (1.02)	–	ATH	C	23.73	5.08	ICISG	8
Howell et al. [19]	95 (53.68%)	69.47	17.59 (3.11)	– (–)	–	ATH	B	41.05	14.74	ICISG	8
Howell et al. [50]	40 (50.00%)	90.00	15.45 (1.15)	– (–)	–	ATH	HS	37.50	12.50	ICISG	7
Hutchison et al. [78]	54 (33.33%)	67.00	19.78 (1.89)	– (–)	64.00	ATH	C	38.89	6.94	NR	9
Iverson et al. [86]	97 (42.27%)	68.04	17.26 (2.00)	– (–)	–	ATH	B	37.11	–	AAN	9
Louey et al. [79]	289 (10.03%)	100.00	25.40 (4.59)	– (–)	–	ATH	O	0.00	0.00	ICISG	8
Mayr et al. [51]	32 (50.00%)	81.25	21.60 (2.50)	– (–)	–	GEN	–	–	–	ICISG	8
McCrea et al. [54]	56 (50.00%)	100.00	– (–)	– (–)	–	ATH	HS	100.00	0.00	AAN	9
Nelson et al. [80]	162 (54.94%)	45.06	29.65 (7.58)	– (–)	32.91	GEN	–	–	–	DoD	7
Nelson et al. [55]	174 (54.02%)	55.17	29.82 (7.69)	0.40 (0.68)	32.34	GEN	–	–	–	DoD	7
Pearce et al. [81]	23 (34.78%)	100.00	25.10 (4.50)	– (–)	–	ATH	O	0.00	0.00	NR	6
Ponsford et al. [87]	223 (55.16%)	69.51	35.17 (12.05)	– (–)	–	GEN	–	–	–	NR	7
Radoi et al. [100]	62 (54.84%)	64.52	– (–)	– (–)	–	GEN	–	–	–	WHO	6
Register-Mihalik et al. [82]	170 (77.65%)	72.94	19.05 (1.21)	– (–)	–	ATH	C	–	–	NR	6
Resch et al. [6]	80 (50.00%)	72.50	19.60 (1.27)	0.28 (0.62)	40.00	ATH	C	70.00	5.00	AAN	8
Rieger et al. [88]	69 (56.52%)	68.12	13.47 (2.52)	0.00 (0.00)	0.00	GEN	–	–	–	NR	6
Schatz et al. [13]	138 (52.17%)	62.32	16.88 (2.01)	0.00 (0.00)	–	ATH	HS	37.68	12.32	AAN	6
Sosnoff et al. [83]	44 (50.00%)	90.91	19.80 (2.20)	– (–)	59.09	ATH	C	–	–	NR	8
Tsushima et al. [101]	51 (50.98%)	–	15.20 (1.73)	– (–)	–	ATH	HS	100.00	0.00	AAN	8
Mean (SD), cumulative	114.1 (105.4), 3308 (49.88% [13.32%])	78.21	21.91 (3.93)	0.86 (1.01)	35.35	–	–	43.95	5.69	–	7.41

NOS Newcastle–Ottawa Scale. Population: *ATH* athletic, *MIL* military, *GEN* general population. Athlete level: *C* collegiate, *HS* high school, *B* both HS and C, *M* mixed populations, *O* other (club, amateur, and/or professional). Concussion definition used: *ICISG* International Concussion in Sport Group, *AAN* American Academy of Neurology, *DoD* Department of Defense, *NR* not referenced, *WHO* World Health Organization

Table 2 Within-subject study and sample characteristics

Author	Sample size	Sex (% male)	Age mean (SD)	Concussion history mean (SD)	Concussion history (%)	Population	Athlete level	American Football (%)	Soccer (%)	Concussion definition used	NOS
Abbassi and Sirmon-Taylor [66]	30	73.33	16.23 (2.40)	0.80 (0.61)	–	ATH	HS	56.66	3.33	ICISG	8
Asken et al. [102]	84	73.81	20.40 (1.30)	1.48 (0.85)	47.62	ATH	C	65.48	4.76	ICISG	9
Broglio et al. [103]	24	83.00	– (–)	– (–)	–	ATH	C	78.67	10.67	AAN	9
Broglio et al. [67]	21	76.19	19.81 (1.25)	1.76 (2.02)	–	ATH	C	76.19	9.52	ICISG	9
Collie et al. [36][a]	145	100.00	23.19 (3.67)	2.29 (2.24)	–	ATH	O	0.00	0.00	ICISG	6
Covassin et al. [104]	154	58.44	17.50 (2.30)	– (–)	–	ATH	B	41.60	27.30	ICISG	8
Covassin et al. [105]	95	41.05	17.74 (2.22)	– (–)	–	ATH	B	0.00	100.00	ICISG	6
Covassin et al. [73]	598	70.23	17.47 (2.28)	0.92 (1.17)	48.83	ATH	B	–	–	NR	8
Covassin et al. [106]	165	67.27	16.68 (2.37)	0.50 (0.89)	33.90	ATH	B	–	–	ICISG	7
Covassin et al. [74]	122	–	17.60 (2.70)	– (–)	–	ATH	B	–	–	ICISG	8
Covassin et al. [107]	104	70.19	16.41 (2.19)	1.20 (1.26)	–	ATH	B	53.80	26.00	ICISG	7
Covassin et al. [68]	222	70.72	16.78 (1.21)	1.08 (1.29)	–	ATH	B	54.50	18.92	ICISG	8
Covassin et al. [108]	57	54.39	20.75 (1.60)	0.74 (0.97)	36.84	ATH	B	15.79	21.05	AAN	7
Covassin et al. [109]	79	51.90	– (–)	0.84 (0.88)	52.00	ATH	C	11.39	24.10	AAN	6
Day et al. [69]	21	52.38	19.50 (0.70)	– (–)	63.64	ATH	C	–	–	NR	6
Del Rossi [35]	21	95.24	16.00 (1.10)	– (–)	–	ATH	HS	–	–	NR	7
Eckner et al. [18][a]	54	92.31	– (–)	– (–)	–	ATH	C	74.07	7.41	NR	9
Eckner et al. [21]	9	88.89	20.56 (1.01)	1.00 (1.12)	55.56	ATH	C	88.89	11.11	NR	8
Haran et al. [77][a]	528	100.00	29.10 (5.00)	– (–)	–	MIL	–	–	–	DoD	6
Hutchison et al. [78][a]	54	67.00	19.78 (1.89)	– (–)	64.00	ATH	C	38.89	6.94	NR	9
Iverson et al. [86][a]	97	68.04	17.26 (2.00)	– (–)	–	ATH	B	37.11	–	AAN	9
Kontos et al. [70]	138	100.00	15.96 (1.18)	0.47 (0.78)	–	ATH	HS	100.00	0.00	ICISG	7
Kontos et al. [110]	75	68.00	16.84 (1.29)	1.08 (1.29)	–	ATH	B	–	–	ICISG	8
Kontos et al. [111]	96	81.25	19.33 (2.08)	0.91 (1.19)	–	ATH	B	57.29	15.63	AAN	8
Lee et al. [112]	1422	89.87	– (–)	– (–)	–	ATH	M	76.37	3.66	ICISG	8
Louey et al. [79][a]	289	100.00	25.40 (4.59)	– (–)	–	ATH	M	0.00	0.00	ICISG	8
Lovell and Solomon [113]	138	0.00	– (–)	– (–)	21.81	ATH	HS	0.00	0.00	ICISG	8
McClincy et al. [114]	104	87.50	16.11 (2.22)	– (–)	32.70	ATH	B	79.80	5.77	AAN	6
McCrea et al. [54][a]	56	100.00	– (–)	– (–)	–	ATH	HS	100.00	0.00	AAN	9
Merritt and Arnett [115]	55	85.50	19.91 (1.40)	– (–)	49.10	ATH	C	50.90	10.90	ICISG	8
Mihalik et al. [116]	180	84.44	16.51 (3.02)	– (–)	37.80	ATH	HS	72.80	9.40	NR	7
Pearce et al. [81][a]	23	100.00	25.10 (4.50)	– (–)	–	ATH	O	0.00	0.00	NR	6
Register-Mihalik et al. [82][a]	170	72.94	19.05 (1.21)	– (–)	–	ATH	C	–	–	NR	6
Resch et al. [6][a]	80	72.50	19.60 (1.27)	0.28 (0.62)	40.00	ATH	C	70.00	5.00	AAN	8

Table 2 (continued)

Author	Sample size	Sex (% male)	Age mean (SD)	Concussion history mean (SD)	Concussion history (%)	Population	Athlete level	American Football (%)	Soccer (%)	Concussion definition used	NOS
Schatz and Robertshaw [117]	250	73.20	15.80 (1.90)	– (–)	32.00	ATH	B	49.00	13.00	NR	7
Sosnoff et al. [118]	20	80.56	21.21 (1.49)	– (–)	–	ATH	C	–	–	AAN	6
Sosnoff et al. [83][a]	44	90.91	19.80 (2.20)	– (–)	59.09	ATH	C	–	–	NR	9
Sufrinko et al. [119]	265	73.58	17.09 (2.20)	0.86 (1.10)	50.57	ATH	M	–	–	ICISG	7
Van Kampen et al. [120][a]	192	69.27	16.86 (–)	0.26 (0.60)	18.75	ATH	B	43.23	15.63	AAN	8
Zhu et al. [121][a]	19	100.00	20.29 (1.59)	– (–)	–	ATH	C	100.00	0.00	ICISG	8
Zuckerman et al. [122]	80	50.00	15.85 (1.82)	0.67 (1.09)	–	ATH	HS	0.00	100.00	NR	8
Mean (SD), cumulative	155.6 (237.5), 6380	75.85	18.96 (2.09)	0.95 (1.11)	43.78	–	–	49.75	15.52	–	7.56

[a]Study with both between- and within-subject designs and were included separately in the between- and within-statistical models

NOS Newcastle–Ottawa Scale. Population: *ATH* athletic and *MIL* military. Athlete level: *C* collegiate, *HS* high school, *B* both HS and C, *M* mixed populations, *O* amateur or professional. Concussion definition used: *ICISG* International Concussion in Sport Group, *AAN* American Academy of Neurology, *DoD* Department of Defense, *NR* not referenced

[28, 62], which suggests that RT-deficit magnitudes may not have been fully explored in previous publications. Relatively small-sample sizes can also mask true effects [28, 62], making it possible that group differences were not observed at the intermediate term in some individual studies due to insufficient statistical power.

Similar to neurophysiological post-concussion research [58–61], it is unknown at this time what the clinical implications are of post-concussion RT deficits extending beyond typical clinical recovery. Previous research has identified heightened musculoskeletal injury risk for multiple years post-concussion and has been postulated to be related to functional RT [22, 63] and neuromuscular deficits [58, 64]. A recent study among concussed and a healthy-matched cohort of collegiate athletes identified 1.78 greater odds for sustaining a lower extremity musculoskeletal injury post-concussion than in the healthy cohort, but no clinical assessments, including the ImPACT RT domain score and a clinical RT assessment [18], significantly predicted musculoskeletal injury [65]. Future work should aim to establish the relationship between lingering RT deficits and clinical, functional, and neurophysiological outcomes to establish their true clinical meaning.

Our approach to examining RT deficits across time was binning effect sizes into acute (0–3 days), sub-acute (4–17 days), intermediate-term (21–59 days), and long-term (80–365 days) categories based on each study's post-injury assessments. RT deficits were binned rather than kept as continuous due to the time points being heavily skewed to the acute timeframe. Statistically weighting the moderator model would have resulted in a biased mean effect value and would have exaggerated the RT deficits observed. The specific timeframes were selected based on both theoretical periods and data-driven binning. For example, previous research has focused heavily on the < 3 days post-concussion time point [54, 66–70], and a data-driven split occurred between 17 and 21 days post-injury due to no studies examining RT within this timeframe. Overall, medium RT-deficit mean effects were present for both between- and within-subject models acutely (0–3 days), and small-to-medium negative effects sub-acutely (4–17 days). Within-subject effects were restricted to acute and sub-acute time points due to limited effects ($k < 5$) outside these timeframes, which restricts our understanding of RT recovery among within-subject study designs. RT deficits were significantly different between the sub-acute time points for between- and within-subject models, which may have occurred due to numerous confounding factors.

The sub-acute model differences may be attributed to the included populations (between-subject = 49% athletes; within-subject = 96% athletes) as well as inherent differences in comparison groups used. Suboptimal performance, also known as "sandbagging" [71], is a concern among clinicians

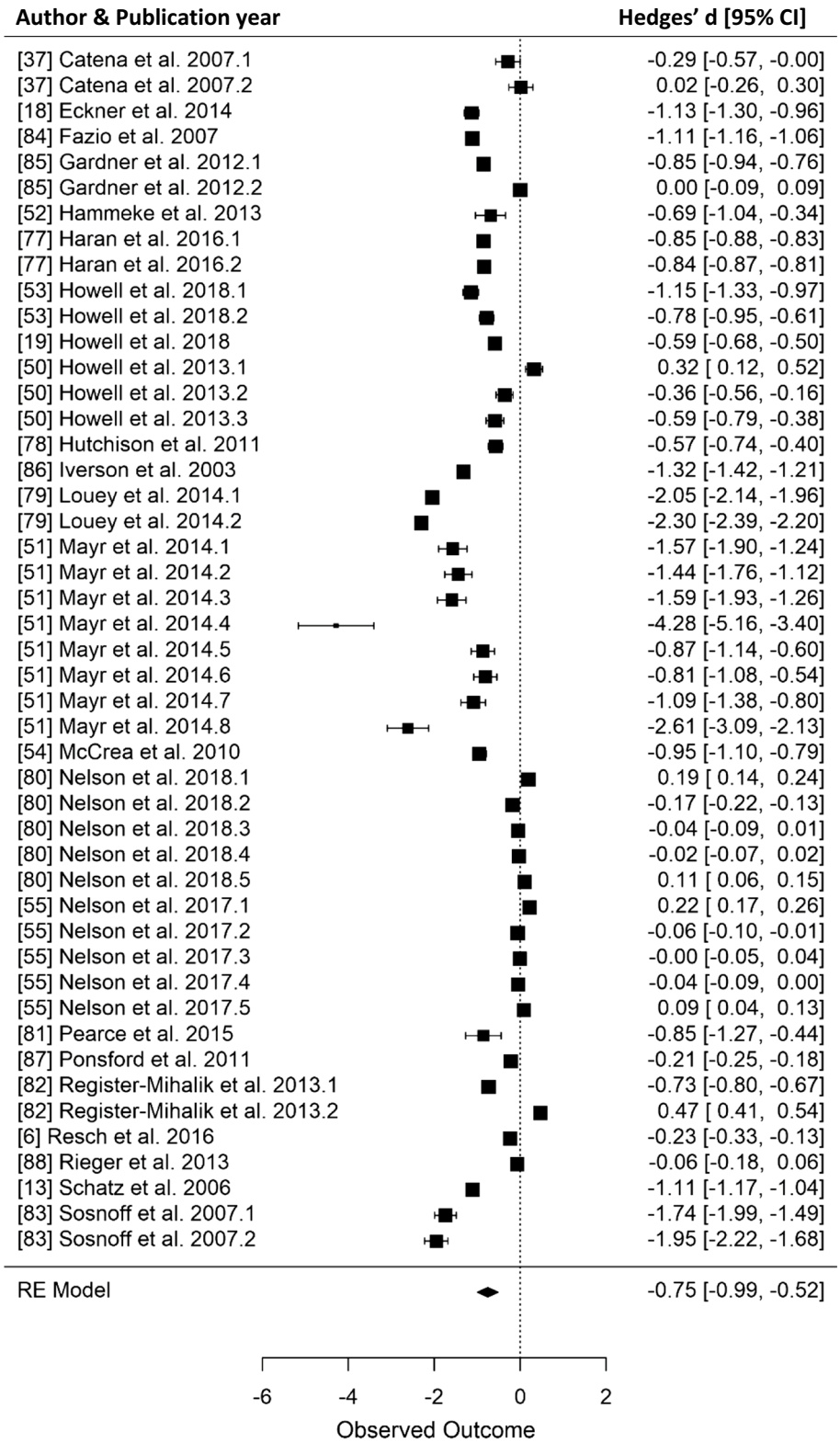

Fig. 2 Between-subject acute (0–3 days) reaction time effects post-concussion. Individual between-subject study mean effects ($k = 47$; Hedges' d) plotted with squares weighted on sample size. 95% CIs are indicated with brackets around each mean effect. The pooled mean effect (center of diamond) with 95% CI (horizontal diamond points in bottom row) using a random-effects model (RE model) with restricted maximum-likelihood estimation. Numerical output provided to the right of each plotted effect is shown. Negative effects indicate increased (i.e., worse) reaction time in the concussed group compared to the control group. Decimal values following first author and publication year indicate the number of multiple effects used from a single study

and researchers who administer concussion baselines due to high effort and motivation needed for thorough, multidimensional baseline assessments. Previous research has shown that sandbagging can successfully occur 11–35% of the time on computerized neurocognitive testing alone [71, 72]. Differences for the between- and within-subject effects may partly be attributed to suboptimal performance given during baseline assessments and could give the illusion that less

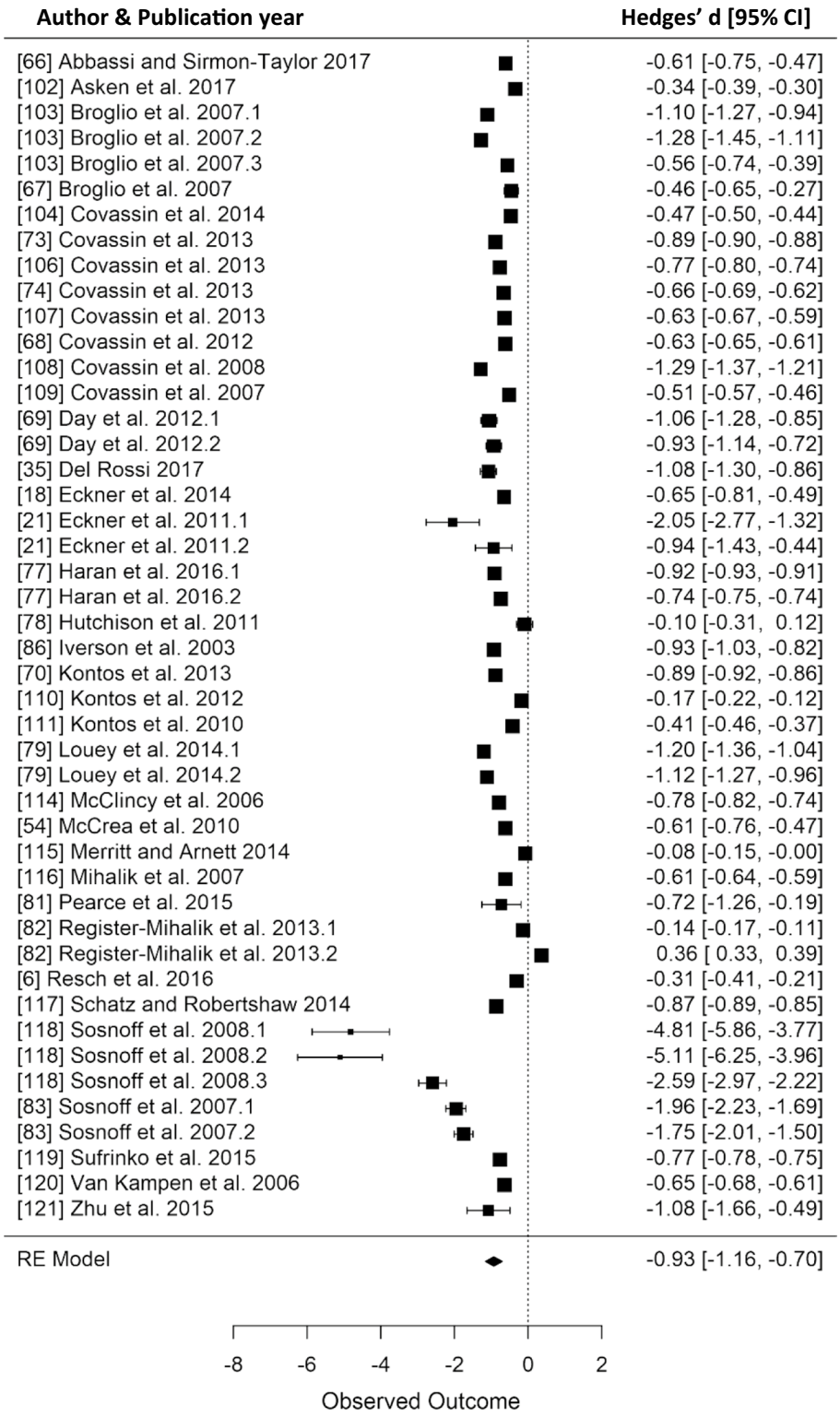

Fig. 3 Within-subject acute (0–3 days) reaction time effects post-concussion. Individual within-subject study mean effects ($k = 46$; Hedges' d) plotted with squares weighted on sample size. 95% CIs are indicated with brackets around each mean effect. The pooled mean effect (center of diamond) with 95% CI (horizontal diamond points in bottom row) using a random-effects model (RE model) with restricted maximum-likelihood estimation. Numerical output provided to the right of each plotted effect is shown. Negative effects indicate increased (i.e., worse) reaction time in the post-injury compared to pre-injury/baseline in a concussed sample. Decimal values following first author and publication year indicate the number of multiple effects used from a single study

deficits and/or quicker recovery have occurred. Sandbagging behavior has not been studied for clinical RT assessments [18, 35] and is an inherent limitation to their use.

Acute post-concussion RT deficits are commonly reported in the literature [13, 18, 19]. However, numerous differences in population and methodological factors have raised concerns among researchers and clinicians as to how robust RT deficits are [15], and what factors should be considered when interpreting outcomes [68, 73, 74]. Our findings suggest fairly robust RT deficits acutely (0–3 days)

Table 3 Relationships between acute (0–3 days) reaction time deficits and potential moderators

	Between-subject study designs						Within-subject study designs					
	k	Mean ES	95% CI	*p*[b]	*Q*[c]	I^2	*k*	Mean ES	95% CI	*p*[b]	*Q*[c]	I^2
Time post-concussion[a]												
Acute (0–3 days)	47	− 0.7279	− 0.9919, − 0.4639	**< 0.0001**	465.4981	88.66	46	− 0.7472	− 0.9089, − 0.5855	**< 0.0001**	295.9764	89.21
Sub-acute (4–17 days)	46	− 0.5655	− 0.6958, − 0.4352	**< 0.0001**	197.6713	61.39	36	− 0.1826	− 0.2648, − 0.1004	**< 0.0001**	103.6718	57.09
Intermediate term (21–59 days)	26	− 0.3219	− 0.5988, − 0.0450	**0.0245**	59.2439	58.45	3	–	–			
Long term (80–365 days)	10	0.3505	− 0.4787, 1.1797	0.3639	81.8114	90.52	0	–	–			
Sex (% male)	47	− 0.0171	− 0.0312, − 0.0029	**0.0193**	291.0144	85.84	45	− 0.0058	− 0.0177, 0.0061	0.3304	282.1132	88.86
Age (mean years)	45	0.0174	− 0.0336, 0.0687	0.4932	398.9722	89.66	40	− 0.0368	− 0.0976, 0.0241	0.2287	282.5347	90.81
Reaction time test type[c]												
Clinical	1	–	–	0.9551	440.6202	89.61	3	–	–	–	–	–
Computerized Neurocognitive testing	29	− 0.7072	− 0.8375, − 0.5769				42	–	–			
Other	17	− 0.7253	− 1.2816, − 0.1690				1	–	–			
Reaction time measure type[d]												
Simple	14	− 0.7754	− 1.0617, − 0.4891	0.5018	276.6511	89.01	14	− 0.9826	− 1.2586, − 0.7066	**0.0438**	170.6570	86.66
Procedural	6	− 0.7176	− 0.8860, − 0.5492				2	–	–			
Mixed	12	− 0.6409	− 0.8712, − 0.4106				25	− 0.6557	− 0.9628, − 0.3486			
CNT Platform[d]												
ANAM	10	− 0.4998	− 0.8222, − 0.1774	0.8633	173.7592	87.93	6	− 0.3735	− 0.7736, 0.0267	**0.0004**	237.7018	85.32
HeadMinder CRI	2	–	–				8	− 1.4799	− 2.0242, − 0.9356			
ImPACT	10	− 0.4784	− 0.7185, − 0.2383				25	− 0.6749	− 1.0906, − 0.2592			
Concussion definition[d]												
AAN	8	− 0.8026	− 1.2871, − 0.3180	0.2930	295.0037	88.25	14	− 0.9919	− 1.3074, − 0.6763	0.1723	280.6444	88.19
DoD	12	− 0.2638	− 1.0766, 0.5490				2	–	–			
ICISG	16	− 1.0922	− 1.8037, − 0.3807				15	− 0.6085	− 1.0119, − 0.2051			
Not referenced	9	− 0.654	− 1.3655, 0.0575				15	− 0.7074	− 1.1174, − 0.2974			
Concussion history (%)	18	− 0.0136	− 0.0398, 0.0126	0.2871	97.1885	92.00	19	− 0.006	− 0.0243, 0.0123	0.4998	68.4564	86.95
Concussion history (mean)	13	0.0576	− 0.5292, 0.6444	0.8330	166.5871	85.77	17	0.203	− 0.1876, 0.5936	0.2853	47.9667	69.80
Population type												
Athlete	23	− 0.8907	− 1.2059, − 0.5755	0.0763	324.0532	86.59	44	–	–	–	–	–
General population	21	− 0.3261	− 0.9351, 0.2829				0	–	–			
Military	2	–	–				2	–	–			
Athlete level[d]												
Mixed	3	–	–	0.8914	84.0242	79.25	13	− 0.7004	− 0.9830, − 0.4178	0.9451	254.2556	89.23
College	9	− 0.7865	− 1.3955, − 0.1775				22	− 0.7667	− 1.1642, − 0.3692			
High school	6	− 0.7337	− 1.3748, − 0.0926				5	− 0.7524	− 1.2940, − 0.2109			
Football (% sample)	21	0.0017	− 0.0084, 0.0118	0.7273	130.2343	81.13	29	0.0007	− 0.0041, 0.0055	0.7613	64.6184	61.52

Table 3 (continued)

	Between-subject study designs						Within-subject study designs					
	k	Mean ES	95% CI	p^b	Q^c	I^2	*k*	Mean ES	95% CI	p^b	Q^c	I^2
Soccer (% sample)	19	0.0084	− 0.0524, 0.0691	0.7747	128.9664	82.57	53	0.0040	− 0.0110, 0.0189	0.5909	60.8548	61.43

All moderator analyses conducted with a mixed-effects multi-level regression model with each individual study serving as the nesting factor

CNT computerized neurocognitive testing, *ANAM* Automated Neuropsychological Assessment Metrics, *HeadMinder CRI* HeadMinder Concussion Resolution Index, *ImPACT* Immediate Post-concussion Assessment and Cognitive Test, *AAN* American Academy of Neurology, *DoD* Department of Defense, *ICISG* International Concussion in Sport Group

[a]Models included all time points; remaining moderator analyses restricted to only acute (0–3 days post-injury) effects

[b]*p* values presented are for each respective omnibus moderator analysis

[c]*Q*-statistics all significant at $p \leq 0.0001$

[d]Only moderators with ≥ 5 effects per category are presented

post-concussion, with medium negative effect sizes for both between- ($D+=-0.7279$) and within-subject study designs ($D+=-0.7472$; Table 3), indicating negligible differences between study designs. These findings may have direct research and clinical implications. Within-subject designs require a pre-injury baseline measure and are traditionally not as feasible outside sports medicine or military settings, and require a large time burden to collect data from a relatively small proportion of post-injury participants. Removing baseline measures may reduce the significant time, financial, and personnel burden associated with conducting thorough pre-injury assessments and could allow optimal resource utilization. Our findings may suggest that baseline measures add minimal clinically important information in post-injury examination compared to normative or control group data specifically for RT, but further diagnostic accuracy and epidemiological work are warranted. Our findings indicating minimal clinical utility with pre-injury baseline measures are in line with recent work examining multidimensional concussion assessment batteries overall [1, 8, 75]. Reaction time is only one component of the recommended assessment battery and our findings alone should not be directly interpreted as a call to forego baseline assessments.

Though we observed fairly consistent RT deficits acutely, our mean effects differ from a previous meta-analysis examining only computerized neurocognitive testing outcomes 0–7 days post-injury [76]. Kontos et al. analyzed 37 RT effects from 37 studies using any computerized neurocognitive testing platform following concussion and observed mixed RT cumulative effects for simple and complex RT measures. In our moderator analyses only examining computerized neurocognitive testing ($D+$ range − 0.3735, − 1.4799) and RT measure types ($D+$ range − 0.6409, − 0.9826), we observed medium-to-large negative effects which still suggest larger magnitude RT deficits post-concussion than the previous work [76]. Differences between their observed mean effect sizes and ours are likely driven by different methods employed. For example, we explicitly aimed to examine all RT outcomes post-concussion which resulted in more effects and studies separately for the between- and within-subject models, included numerous computerized and non-computerized RT test and measure types, and had multiple effects from some studies.

The cumulative and diverse studies examining post-concussion RT allowed for a detailed assessment of potential RT moderators (Table 3). Previous research has indicated that patient characteristics such as age, sex, and concussion history may alter clinical recovery [33]. We similarly observed sex (% male) to be an influential factor for RT deficits for between-subject effects. For every 1% increase of males in the sample, RT deficits worsened by $D+=0.0171$ (Table 3) and indicate males often presented with greater RT deficits post-concussion. Though the sex difference

effect magnitude was statistically significant, the findings may lack clinical value as this finding indicates a 10% increase in male sample would result in a small magnitude RT deficit of $D+=0.1710$. We did not observe significant effect size differences based on age or concussion history for between- or within-subject models. The lack of differences for age and concussion history may be attributed to the fairly homogeneous age ranges among included studies (between: 21.78 ± 3.82 years; within: 18.96 ± 2.09 years) and self-reported nature and variance in previous concussion frequencies among patients when examining concussion history. A recent systematic review indicated nearly 50–50 splits among studies regarding whether younger age or prior concussions result in longer clinical recovery [33]. The large discrepancies in prolonged recovery risk factors between studies may be driven by relatively small-sample sizes used in earlier concussion research coupled with limited effect sizes being reported, resulting in a critical statistic missing for clinicians and researchers to understand the magnitude of risk factors. Though we did not examine all clinical outcome measures for determining clinical recovery after concussion, our findings may provide insight into the mixed evidence and suggest age and concussion history minimally influence RT deficits.

Variance in RT assessment and measure types, populations, concussion definition, athlete level, and sport type may alter outcomes. Our analytic approach addressed these methodological concerns, with overall findings suggesting only the RT measure type and computerized neurocognitive testing platform significantly influenced RT deficits. Significant RT measure type and computerized neurocognitive testing platform differences were only observed in the within-subject effects. Surprisingly, simple RT measures demonstrated larger magnitude RT deficits following concussion than mixed measure types. These 14 simple RT effects [18, 19, 37, 54, 55, 77–83] incorporated tasks that required pressing a single computer button or pressing a trigger in response to a stimulus, whereas the 25 mixed RT effects [6, 13, 52, 53, 55, 79, 80, 84–88] were composite scores derived from subtests within computerized neurocognitive tests such as ImPACT and CNS Vital Signs.

Previous research has suggested that more cognitively demanding RT assessments such as procedural RT may elicit greater deficits [38, 49, 58], but this was not observed among our cumulative findings. At the crux of simple RT, assessments in healthy individuals are the delays between detecting either auditory (57 ms latency) or visual (89 ms latency) stimulus [89], transcallosal transmission across brain hemispheres (3 ms latency) [90], and motor signaling (25 ms latency; 85–117 ms total latency) [91] to the muscles required to elicit the response traditionally measured in clinical settings via RT. The addition of decision-making, greater cognitive processing, or more gross motor requirements adds additional latency due to the increased neurophysiological demands [89]. Though we observed greater RT deficits for simple RT than mixed, these differences may be attributed to comparing the varying simple RT assessments to the fairly standard computerized neurocognitive test mixed RT types. The 95% CIs for simple and mixed RT do however overlap and may suggest minimal, clinically unimportant differences between the measure types. Regardless, clinicians and researchers should continue incorporating outcomes ranging in complexity and primary psychometric domains assessed until a gold standard for concussion diagnosis is established [1, 6, 92].

Computerized neurocognitive testing platforms also resulted in different effect size magnitudes, with HeadMinder CRI demonstrating the largest mean effect (Table 3). Approximately 84% of sports medicine clinicians utilize ImPACT as their computerized neurocognitive assessment [9], which indicated a medium decreased RT effect in our present findings. Clinicians and researchers are often faced with the challenge of selecting a computerized neurocognitive testing platform which vary in their cost, services, and underlying psychometric properties [15]. Though differences in cumulative effect magnitudes were present between RT measure types and computerized neurocognitive testing platforms, all outcomes produced medium-to-large negative effects. Clinically, these findings suggest that certain tools may elicit larger deficits compared to others, but post-concussion deficits are present acutely regardless of tool use.

Differences between concussion definitions employed among studies were not statistically significant for between- or within-subject effects (Table 3), suggesting that concussion definitions are a minimal consideration when interpreting RT findings. Differences between athletic and general public populations were not statistically different from each other for between-subject effects (Table 3). However, the moderator analysis neared a priori statistical significance ($p=0.0763$) with general population studies having inconsistent acute RT deficits ($D+=-0.3261$, 95% CI -0.9351, 0.2829) relative to athletic settings ($D+=-0.8907$, 95% CI -1.2059, -0.5755). We were not able to examine population differences among within-subject study designs as limited studies used military or general public populations. We noted multiple general population [55, 80, 87, 88] studies utilizing a concussed vs orthopedic injured control group typically recruited from emergency departments during data extraction. The between-subjects forest plot (Fig. 2) also indicated numerous effects and 95% CIs crossing zero, with the majority (66%; 6/9) being studies employing a concussed vs. orthopedic injured control comparison. We did not employ a moderator analysis as this was not an a priori analysis; however, this observation suggests that utilizing orthopedic injured control data may not be the best "control" group. Neurocognitive testing deficits are present after any

type of traumatic injury to the body and may be attributed to psychological and emotional responses associated with a traumatic and pain-related orthopedic injury. Researchers and clinicians should avoid solely comparing concussed individuals to orthopedic injured individuals as orthopedic injuries often mimic typical post-concussion neuropsychological outcomes observed [93, 94].

Our study has inherent limitations worth noting. First, our findings are limited to the available peer-reviewed and published research and those meeting our inclusion criteria. The majority of included studies were in high-school or college aged individuals in athletic and military settings (Tables 1, 2), and suggests that our findings may not translate to populations outside the examined work. Only 9 of 29 (31.0%) author groups missing statistical output to calculate effect sizes provided the requested information for meta-analysis, further limiting the literature available and potentially introducing potential bias. However, this is in line with previous meta-analyses and is unlikely to bias our outcomes given the large sample. Second, concussion diagnoses currently lack a gold standard diagnostic assessment which likely introduces large variance into the populations examined. These individual studies, and thus our findings, are derived from imperfect diagnosed concussions and may influence the mean effects observed. Third, wide test–retest reliability variation for each RT assessment type is reported in the literature [15, 95–97]. Inconsistent measurement across timepoints is a confounding aspect for all RT outcomes and should be considered based on (1) the RT assessment being used and (2) the sample which it is employed in. Finally, we observed significant funnel plot asymmetry and a few outliers on *Q*–*Q* plots (Electronic Supplementary Material Figure S1) which suggests skewed effect sizes and potential publication bias among these data. However, funnel plots are not specific to publication bias [48, 98]. These potential asymmetry concerns, however, may also highlight (1) fairly robust RT deficits acutely following concussion and (2) the heterogeneous nature of concussion outcomes (Figs. 2, 3).

5 Conclusion

Post-concussion RT deficits are large and robust acutely, moderately present at the intermediate term, and diminish at the long term. Study design (between-subject vs within-subject), sex, RT measure type, and computerized neurocognitive testing platform statistically influence RT deficits, but likely not to the degree that represent major clinical considerations when assessing post-concussion impairments. Clinicians should consider their time and resources available, and current best practices when implementing concussion assessment tools. However, RT should be incorporated within the multi-faceted assessment battery acutely and during the return to play process.

Data Availability Statement All data generated or analyzed during this study are included in this published article (and its Electronic Supplementary Material).

Compliance with Ethical Standards

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Conflict of Interest Landon Lempke, David Howell, James Eckner, and Robert Lynall declare that they have no conflicts of interest relevant to the content of this review.

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